Benchmarking Counterfactual Explanations of Model Change with Exact Ground Truth

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Abstract. Without ground truth one cannot tell which kind of change each method for explaining a model update detects. We contribute a benchmark with exact ground truth and a linear analysis of what five families of explanations see, among them differential counterfactuals (Δ-CE): the nearest inputs on which exactly one model changes its decision, both by a margin. On small CNNs, though not on a ResNet-8, gradient differences miss a threshold shift and see a rotation, as predicted. On gently fine-tuned null updates Δ-CE finds no witness, also through a learned autoencoder, while counterfactual and attribution differences report a change; with two independent changes, witnesses point to the one near the query. A pre-specified hypothesis that Δ-CE names the changed factor more often than gradient differences was rejected. This abstention comes from the margin check, which can also filter the counterfactual of the new model; the Δ-CE search adds witnesses that such filtering misses.

1 Introduction

An update that improves accuracy can still change individual decisions in unwanted ways [1]. Explaining *how* the models differ is a goal of explainable AI under distributional change [2, 3], distinct from explaining a shift of the data [4]. Artelt et al. [5, 6] compare the counterfactuals [7] of one input under each model, or take a difference of gradients; Muschalik et al. [8] compare feature importances; disagreement search [9, 10, 11] looks for inputs on which the models disagree. Ground-truth benchmarks exist for explaining one model [12, 13, 14], not a change, so what each family can see goes unchecked.

We say that a method *sees* a change if it stays silent when nothing changed (abstention), reports a change when there is one (*whether*), and names the changed factor and region (*what*, *where*). As a fifth family we study *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision, both models by a margin κ. Unlike robust counterfactuals [15, 16], valid for both models, it is valid for exactly one.

Contributions. (i) A benchmark with exact ground truth: five rendered factors, four shift types, null updates and a 4×4 grid over rotation and threshold. (ii) A linear analysis separating rotation from offset, tested on CNNs and a ResNet-8 under a protocol fixed in advance; all failures are reported. (iii) Δ-CE stays silent on gently fine-tuned null updates where counterfactual and attribution differences report change, also on unscreened nulls and through a learned decoder; it localises two independent changes, and its search finds witnesses where the filtered counterfactual of f_{new} finds none.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For a query z_0 that both models assign to class y, the margin is the logit difference $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$; z_0 is valid if $m_o(z_0), m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|_2$ if $\|z^\star - z_0\|_2 \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models disagree, both decisively, unlike a difference of counterfactuals.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness. *Proof:* subtract the constraints of $\mathcal{F}_\kappa^N$.

Corollary 1 (constant logit offset). If f_{new} adds b to the logit of class 1, e.g. the prior correction [17], then $h \equiv \pm b$ and $\nabla h \equiv 0$. Type-N witnesses exist only for the class whose margin b lowers and only if $|b| \geq 2\kappa$, at the points with $\kappa \leq m_o \leq |b| - \kappa$. *Proof:* insert $h \equiv -|b|$ in Lemma 1; no linearity is needed. An offset thus moves one class towards the boundary of f_{new} and the other away.

Linear case. For linear margins on $\mathbb{R}^p$ write the unit normals as $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$, $v \perp \hat{w}_o$ ($\theta = 0$: offset; $\theta > 0$: rotation towards v, which f_{old} ignores). Let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$, and u_n, u_o the steps to the minimum-norm counterfactuals of f_{new}, f_{old}. The three explanations are the gradient difference [5], $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing, $u_n - u_o$, and the witness, $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference has a larger component along v than along $\hat{w}_o$ for every θ, and so does the witness iff $\theta > \pi/4$ or $d_o(\sin\theta + \cos\theta) < d_n$. After projecting out $\hat{w}_o$, all three coincide up to positive scale, as in Theorem 1 of [5] (new here: the offset case and the witness). *Proof* of (a): both normals equal $\hat{w}_o$; of (b): online supplement. In the benchmark, a higher label threshold is an offset and a rule that adds colour a rotation.

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0,1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise. The exact renderer gives the factor change of any explanation without a learned oracle, and all methods search the same factor space through it. f_{old} is a small CNN (five seeds); f_{new} is f_{old} fine-tuned on 10 000 samples from one condition (Table 1). A pair enters only if its change is visible in decisions on validation data. The ground truth is the injected change; what f_{new} learns is read from the realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of a logistic fit to its decisions, at which the grid is analysed.

Table 1: Adaptation conditions; $k^\star$: ground-truth factor.

Change	Shift	Adaptation data	$k^\star$	Pairs
Shortcut	spurious	colour correlated with label (0.99)	colour	3
Subgroup	local concept	labels flipped for size > 0.75	size	3
Threshold	concept	$y = \mathbb{1}[\text{aspect} > 0.6]$	aspect	3
Null	none / covariate	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid[a]	concept	$y = \mathbb{1}[\text{aspect} + \alpha(\text{colour} - 0.5) > t]$	aspect / colour	8 / 24
Offsets[b]	concept	grid, $\alpha = 0$, $t \in \{.53, .56, .6\}$, seeds 2–4	aspect	9
Both[c]	spurious + local	Shortcut and Subgroup at once	colour, size	8

[a] $\alpha \in \{0, .1, .25, .5\} \times t \in \{.5, .53, .56, .6\}$, 2 seeds; 4 offsets tested. [b] 8 qualified. [b,c] Registered.

Metrics and methods. On $N = 300$ valid test points per pair, Recovery@r is the share with a valid explanation within the budget r whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0); the found-rate is the share with any valid explanation within r. For Δ-CE we count type-N witnesses only (pre-declared), so a query whose class the update raises counts 0. All counterfactual methods share one solver, the test points and the budget $r = 0.54$, set once on validation. A margin score, $|h(z_0)|$, serves as the control for *whether*. We compare Δ-CE ($\kappa = 0.5$) with **CF-diff**, counterfactual differencing [5]; **Grad-diff**, the gradient difference [5]; **Imp-diff**, a difference of permutation importances [8], one vector per pair; **Seg-search**, a line search towards training points, after [9]; **No-prox**, Δ-CE without proximity, the DeepXplore objective [10], to which DiG-IN [11] corresponds on an exact manifold; **CF-new**, the counterfactual of f_{new}; and, in Section 4.2, IG- [18] and SHAP-differences [19].

Protocol. Each test set served at most one confirmatory test, after all choices were fixed. The main test set carries the comparison of Table 2. Hypotheses formed after it were tested on new data, and three follow-ups were registered before their data existed [20]: unscreened nulls, offsets and a ResNet-8; the learned decoder; and two changes at once. Bootstrap intervals resample points within pairs, so Table 3 also counts pairs; the supplement repeats every test with the pair as unit and lists every test set with its date. A digit-image tier and a label-shift update failed qualification.

4 Results

4.1 Abstention, detection and localisation

Abstention. No null query has a witness: on the 8 null pairs of Table 2, five null updates of an MLP on Adult, and 8 registered unscreened null pairs, admitted without the qualification check. On the same nulls (covariate-shifted included) CF-diff reports a change for 97.7–100% of points. Silence does not need the exact renderer: searching through the decoder of an autoencoder trained on the images (registered), Δ-CE finds no witness for a null query (0/800; CF-diff 85–94%), and on 6 of 8 offsets finds one for at least half of the lowered-class queries.

With $\kappa = 0$, witnesses appear on 83% of null queries. Silence also depends on the recipe: the nulls are fine-tuned more gently than the offsets (fewer epochs, lower learning rate; supplement), though their threshold moves as far

Table 2: Recovery on the main test set (↑) and false reports on null updates (↓).

Condition	Δ-CE	CF-diff	Grad-diff	Imp-diff	Seg-search	No-prox	CF-new
Shortcut ↑	0.38 (0.84)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
Subgroup ↑	0.53 (0.81)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
Threshold ↑	0.47 (0.47)	0.76 (0.90)	0.07 (1.0)	**1.00**[a]	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null ↓	**0.000**	0.994	1.0	1.0	**0.000**	**0.000**	0.996

Change rows: Recovery@r (found-rate), mean of 3 pairs, CIs ±0.035. Null: found-rate on 8 null pairs; any output is a false report. Bold: best in row. [a]Ranks aspect first in every pair.

Table 3: Tests and their outcomes (✓ confirmed, × not confirmed).

Claim	Data	Estimate	p	Pairs	Status	Set
Abstention, detection and localisation						
No witness on nulls[d]	Nulls, new data; Adult MLP	0/2 400; 0/1 500	–	8/8; 5/5	✓	post
No witness on unscreened nulls	Unscreened nulls	0/2 400	–	8/8	✓	OSF
Same, search via VAE decoder	Nulls, learned decoder	0/800	–	8/8	✓	OSF
No witness on nulls	ResNet-8, $\kappa = 0.5$¶; $\kappa^\star = 2.5$	180/200; 0/200	–	0/2; 2/2	×; expl.	OSF; post
CF-diff (no cut) flags $\lvert b\rvert < 2\kappa$ [d]	Logit offsets	0.961 [0.955, 0.967]	$2{\cdot}10^{-4}$ §	–	✓	post
Witness ≥ margin score +0.2	Registered offsets	+0.17 to +0.97	–	7/8	×	OSF
Witness localises ($L > 0$)	Two changes at once	$L = 0.50$	0.004 (sign)	8/8	✓	OSF
Offsets						
Grad-diff Recovery < 0.2	Grid, $\alpha = 0$	0.081 [0.066, 0.096]	$4{\cdot}10^{-4}$ ‡	4/4	✓‖	pre
Grad-diff Recovery < 0.2	Registered offsets	0.033–0.093	0.004 (sign)	8/8	✓	OSF
Grad-diff unchanged by offset	Grid	−0.098 [−0.143, −0.053]†	0.94 ‡	–	×	pre
Δ-CE > Grad-diff [d]	Threshold, new data	+0.31 [0.27, 0.35]	$2{\cdot}10^{-4}$ *	3/3	✓‖	post
Rotations						
Grad-diff rises with α	Grid	+0.57 [0.53, 0.61]	$4{\cdot}10^{-4}$ ‡	$\rho = 0.89$	✓	pre
Grad-diff Rec. < 0.2; rot. > offset	ResNet-8	0.32–0.36; 0.06–0.10	–	0/3; 0/3	×	OSF
Proj. Δ-CE ≈ Grad-diff	Shortcut, new data; grid	−0.30; −0.35 (CIs ±0.03)†	1 *; 1 ‡	–	×	post; pre
Normal at $z^\star$ halves the gap	Grid, new data	−0.19 → +0.03 [0.01, 0.05]†	$6{\cdot}10^{-4}$ §	4/4	✓‖	post
Δ-*CE versus gradient differences*						
Δ-CE > Grad-diff	Subgroup	−0.09 [−0.13, −0.05]	–	0/3	× rejected	pre
Δ-CE ≈ Grad-diff, CF-diff	Shortcut, Threshold	$\lvert\Delta\rvert = 0.17$–$0.53$	–	–	×	pre

Estimate: null queries with a witness (count), a Recovery or found-rate, or a difference Δ of Recoveries (≈, unchanged: $\lvert\Delta\rvert \le 0.1$); 95% CIs (†90%); –: not applicable. Pairs: model pairs on the claimed side / all; ρ: rank correlation over 24 pairs; ‖confirmed at point level only (≤ 4 pairs, pair-level $p \ge 0.0625$). p: Holm-adjusted within families *, ‡, §. Set, when the test was fixed: OSF = registered [20] before its data existed; pre = before any result, or before the grid run; post = after earlier results, then tested on new data. [d]Follows from a lemma or the definition, not a risky prediction. ¶Run on a local CPU after its cost rule excluded it; expl. = exploratory; Proj.: $\hat{w}_o$ projected out.

($\lvert\Delta\hat{t}\rvert \le 0.035$); the grid's null cell, fine-tuned like the offsets, has witnesses on 22% of queries at $\kappa = 0.5$ and none at $\kappa = 1$, where small offsets keep few witnesses (0.04). Silence is thus a property of κ relative to the size of the update, as Lemma 1 states, and not of the search.

Detection. With each score cut on the nulls to flag none, witnesses exist for 0.17–1.00 of lowered-class queries on the 8 registered offsets, the margin score flags 0.00–0.54 and CF-diff none. The witness beat the margin score by 0.2 in 7 of 8 pairs, short of the registered 8/8, so we claim no advantage for *whether*.

Localisation. With a colour shortcut everywhere and a label flip for large objects (registered), witnesses change size most for 46–75% of queries near the large-object region (size ≥ 0.60) and 13–23% far from it (size ≤ 0.45); this difference L is positive in all 8 pairs (pooled 0.50; Table 3). Grad-diff and CF-diff barely localise ($L = 0.09$ and 0.13), and Imp-diff, one vector per pair, cannot. CF-new localises as well as Δ-CE (0.51).

Search versus filtering (post hoc, saved outputs). The counterfactual of f_{new} kept only where $m_o \ge \kappa$ (filtered CF-new) is a witness; it is silent on the nulls and localises ($L = 0.48$), but finds witnesses for only 0.42–0.46 of

the main-test queries, against 0.47–0.84 for Δ-CE. The extra witnesses carry Δ-CE's Recovery on Shortcut (0.38 vs. 0.03) and part on Subgroup (0.53 vs. 0.30), higher in 3 of 3 pairs each, but not on Threshold (0.47 vs. 0.45). Null-adjusted (minus the rate of naming $k^\star$ on nulls), Δ-CE leads on Subgroup and Threshold, Grad-diff on Shortcut (0.79); unfiltered CF-new falls to −0.09 on Threshold. The search has a price: a Δ-CE query takes 2.4 s, against 0.6 s for CF-new and under 1 ms for Grad-diff (supplement).

4.2 Offsets, rotations and further baselines

Gradient differences are blind to an offset, as Proposition 2(a) predicts: their Recovery is 0.081 on the grid's pure offsets and 0.03–0.09 on the 8 registered offsets (Table 3). On Threshold, Δ-CE beats them on new data (0.41 vs. 0.10; main test set: 0.47 vs. 0.07) in 3 of 3 pairs, too few for a pair-level test. Type-N witnesses exist only for the class the offset lowers (class 1), hence the found-rate of 0.47 for Δ-CE. Type-O witnesses cover the other class, and the two types together reach Recovery 1.00 on a further test set.

Gradient differences see a rotation, increasingly with its angle: Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$ ($t = 0.5$; Spearman $\rho = 0.89$ over the 24 rotation pairs; map of all pairs in the supplement). The projection equivalence of Proposition 2(b) holds only with $\hat{w}_o$ taken at the witness, not as first tested (Table 3).

Attribution differences and other baselines. With predictions fixed beforehand, IG- and SHAP-differences recover colour on Shortcut (0.93) like Grad-diff and, against the predicted blindness, name aspect on Threshold (0.73–0.75 vs. 0.07), but also on 57% and 71% of null queries (IG-diff on 42% of the registered nulls, as predicted), so this is no signal of change. Imp-diff computed on test data ranks aspect first in all 9 change pairs; on adaptation data it ranks $k^\star$ first. Seg-search with ten times the segments finds witnesses for 0.26–0.47 of queries (predicted ≤ 0.2), none that Δ-CE misses.

Second architecture. On a registered ResNet-8, null fine-tuning moves margins four times more than on the CNNs, and witnesses appear on 0.90 of null queries: κ is model-specific. With κ calibrated on validation nulls (rule fixed in advance; grid extended after $\kappa \leq 2$ failed), $\kappa^\star = 2.5$, no new null query has a witness (0/200) and witnesses exist for 0.78–1.00 of lowered-class offset queries (Δ-CE Recovery 0.36–0.40). Grad-diff sees neither change: on 2 null, 3 offset and 4 rotation pairs it names aspect on 0.32–0.36 of offset queries, near its 0.20–0.30 on nulls, and its rotation Recovery, 0.06–0.10, is below the offset of equal seed and threshold in all 3 comparable pairs.

5 Discussion and limitations

Our results suggest that an explanation of change needs null updates as a reference, since counterfactual and attribution differences report change on them; on small CNNs, gradient differences also miss offsets. Abstention comes from the check $m_o \geq \kappa$, which can also filter the counterfactual of f_{new}; specific to

Δ-CE is the search for witnesses at queries where that counterfactual fails the check. **Limitations.** (1) The data are synthetic and binary; Recovery through the learned decoder is approximate and lower for Δ-CE than for CF-diff; multi-class tasks, other domains, retraining from scratch and architecture changes are untested, and we did not test why the linear prediction fails on the ResNet-8; the tabular test covers abstention only. (2) κ is in logit units, exact only for a constant offset, and must be calibrated on null updates of each model family and fine-tuning recipe, obtained by fine-tuning f_{old} on resampled data. (3) Witnesses see only decisive changes on the manifold within the budget, without a certificate. (4) Several tests were formed after earlier results; changes were logged before their data existed, with three disclosed exceptions. Online supplement and code: github.com/Johnlrr/explaining-model-change.

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